

PIKHTOVKA, Russian Federation

WMO: 295340

Lat: 55.983N

Lon: 82.700E

Elev: 127

StdP: 99.81

Time Zone: 7.00 (E07)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-33.6	-39.4	0.1	-37.3	-35.8	0.1	-33.0	14.2	-12.0	12.4	-12.0	1.5	220	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	27.4	20.2	26.1	19.4	24.6	18.6	22.1	26.1	20.8	24.6	19.6	23.4	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.6	24.7	19.3	14.3	23.5	18.1	13.2	22.1	65.8	26.2	61.0	24.7	56.8	23.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.2	-41.1	29.8	4.6	1.7	-44.4	31.0	-47.1	31.9	-49.7	32.9	-53.1	34.1		
(5)			-41.0	24.5	4.3	2.1	-44.1	26.0	-46.7	27.2	-49.1	28.4	-52.3	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-16.8	-16.5	-9.1	0.5	9.1	14.4	18.0	15.0	8.7	1.6	-9.0	-15.0		
	DBStd	13.80	7.72	7.14	6.23	5.96	4.99	4.34	3.05	3.05	3.79	4.59	7.59	9.17		
	HDD10.0	4225	831	743	592	287	81	12	1	3	69	261	570	776		
	HDD18.3	6680	1089	976	850	534	289	128	42	109	290	518	820	1034		
	CDD10.0	637	0	0	0	2	51	145	249	159	29	1	0	0		
	CDD18.3	52	0	0	0	0	1	11	33	7	0	0	0	0		
	CDH23.3	578	0	0	0	1	41	138	300	89	11	0	0	0		
	CDH26.7	81	0	0	0	0	3	21	50	7	0	0	0	0		
	WSAvg	3.5	3.9	4.0	3.6	4.0	3.8	2.8	2.3	2.3	2.9	3.7	4.3	4.2		
	PrecAvg	462	24	18	20	26	34	57	69	57	41	43	39	35		
(15) Precipitation	PrecMax	652	46	38	42	51	75	120	115	136	93	80	68	58		
	PrecMin	324	5	1	3	6	2	19	13	10	14	14	22	15		
	PrecStd	81	11	9	10	12	20	26	29	26	19	17	14	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	0.4	5.7	19.9	27.0	29.1	30.6	27.3	25.0	16.0	4.3	1.5		
		MCWB	-1.4	-0.5	2.7	11.7	15.1	21.9	22.7	19.6	16.1	10.6	1.7	0.6		
	2%	DB	-2.4	-1.4	3.1	15.6	24.2	26.9	27.9	25.8	21.2	12.7	2.0	-0.3		
		MCWB	-3.2	-2.5	0.4	9.0	14.1	19.8	21.0	19.1	15.6	8.5	0.3	-0.9		
	5%	DB	-4.0	-3.4	1.4	11.9	21.5	25.0	26.5	23.8	18.3	10.3	1.0	-2.2		
		MCWB	-4.6	-4.4	-0.5	6.3	13.3	19.7	20.5	18.6	13.0	6.9	0.1	-2.8		
	10%	DB	-6.0	-5.5	-0.1	8.7	18.9	23.0	25.0	22.2	15.8	8.2	-0.1	-3.7		
		MCWB	-6.7	-6.4	-1.8	4.5	12.3	17.4	19.7	17.7	11.8	5.9	-0.9	-4.4		
	0.4%	WB	-1.3	-0.4	3.4	13.0	17.7	24.1	24.2	22.5	17.6	11.2	2.3	1.0		
		MCDB	-1.0	0.1	5.2	18.9	23.3	25.8	27.4	25.8	22.4	14.8	3.2	1.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.1	-2.4	1.0	9.4	15.6	21.5	22.5	20.6	15.7	9.2	1.0	-0.8		
		MCDB	-2.5	-1.8	2.5	14.6	22.1	25.7	26.8	23.5	21.4	11.8	1.7	-0.2		
	5%	WB	-4.7	-4.4	-0.2	6.7	14.3	20.0	21.4	19.2	13.9	7.5	0.1	-2.6		
		MCDB	-4.0	-3.3	1.1	10.5	20.1	24.0	25.4	23.1	17.3	10.0	0.8	-2.1		
	10%	WB	-6.8	-6.4	-1.7	5.0	12.7	18.3	20.2	17.9	12.2	5.7	-0.9	-4.4		
		MCDB	-6.1	-5.5	-0.3	8.6	18.2	21.8	24.1	21.4	15.2	7.9	0.0	-3.8		
	5% DB	MDBR	9.4	10.5	11.8	9.6	11.5	10.9	10.8	11.4	10.6	7.5	7.3	8.3		
		MCDBR	7.2	8.8	10.9	14.4	17.2	14.1	13.7	14.5	15.5	12.5	7.1	8.6		
		MCWBR	6.1	8.0	9.1	9.0	9.7	9.1	7.9	8.6	9.7	8.3	6.7	8.2		
		MCWBR	6.4	8.2	8.5	13.6	14.9	12.8	12.7	12.6	14.6	11.4	7.0	8.6		
(35) Clear-Sky Solar Irradiance	taub	0.289	0.323	0.372	0.405	0.405	0.408	0.439	0.409	0.369	0.351	0.318	0.291			
	taud	2.005	1.974	1.915	2.069	2.220	2.289	2.200	2.274	2.334	2.244	2.062	1.955			
	Ebn at Noon	568	700	758	804	834	836	797	798	780	680	525	446			
	Edh at Noon	67	103	141	141	129	122	131	115	94	79	64	55			
(44) All-Sky Solar Radiation	RadAvg	0.84	1.85	3.32	4.52	5.20	5.85	5.28	4.33	2.89	1.46	0.70	0.51			
	RadStd	0.08	0.11	0.20	0.34	0.52	0.59	0.38	0.34	0.37	0.15	0.12	0.09			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page